

Whole Exome Sequencing Analysis

Patient name	:	XX	PIN	:	XX
Gender/ Age	:	Male / 23 Years	Sample number	:	XX
Referring Clinician	:	XX	Sample collection date	:	NA
Specimen	:	Peripheral Blood	Sample received date	:	NA
			Report date	:	NA

Clinical history

Proband, XX was born to non-consanguineous parents. XX is presented with chief complaints of hunch on the back, strabismus, long arms, legs, fingers, and toes (arachnodactyly), flat feet, crowded teeth, slight knock-knees and vitamin B12 deficiency. XX height is 6.5 feet and he is unmarried. XX father is 5.11 feet tall, has complaints of hunch on the back and strabismus. XX mother is 5.9 feet tall, has complaints of glaucoma. Proband, XX is suspected to be affected with Marfan syndrome and has been evaluated for pathogenic variations.

Results

Likely pathogenic variant causative of the reported phenotype was identified

List of significant variant identified related to the phenotype:

Gene	Region	Variant*	Allele Status	Disease	Classification*	Inheritance pattern
<i>FBN1</i> (-)	Exon 41	c.5048G>A (p.Gly1683Glu)	Heterozygous	Acromicric dysplasia (OMIM#102370) Geleophysic dysplasia 2 (OMIM#614185) Ectopia lentis, familial (OMIM# 129600) Marfan lipodystrophy syndrome (OMIM#616914) Marfan syndrome (OMIM#154700) Stiff skin syndrome (OMIM#184900) Weill-Marchesani syndrome 2, dominant (OMIM#608328) MASS syndrome (OMIM#604308)	Likely pathogenic (PM1, PM2, PP2, PP3)	Autosomal Dominant

*Genetic test results are based on the recommendation of American college of Medical Genetics [1].

No other variant that warrants to be reported for the given clinical indication was identified

CNV Findings

No significant Copy Number Variations (CNV) related to phenotype was detected.

Interpretation

***FBN1*: c.5048G>A**

Variant summary: A Heterozygous missense variation in exon 41 of the *FBN1* gene (chr15:g.48463916C>T, NM_000138.5, Depth: 94x) that results in the amino acid substitution of Glutamine for Glycine at codon 1683 (p.Gly1683Glu) was detected.

Population frequency: This variant has not been reported in gnomAD database and 1000 genomes databases.

Clinical evidence: This variant has been classified as uncertain significance in ClinVar database [3].

In-silico prediction: The *in-silico* predictions of the variant are damaging by SIFT, PolyPhen-2 (HumDiv) and deleterious by CADD and MutationTaster2. The reference codon is conserved across PhyloP and GERP++ tools.

OMIM phenotype: Acromicric dysplasia (OMIM#102370) and Geleophysic dysplasia 2 (OMIM#614185) are caused by heterozygous mutation in exon 41 or 42 of the *FBN1* gene (OMIM*134797). Acromicric dysplasia (ACMICD) is an autosomal dominant disorder characterized by severe short stature, short hands and feet, joint limitations, and skin thickening. Radiologic features include delayed bone age, cone-shaped epiphyses, shortened long tubular bones, and ovoid vertebral bodies. Affected individuals have distinct facial features, including round face, well-defined eyebrows, long eyelashes, bulbous nose with anteverted nostrils, long and prominent philtrum, and thick lips with a small mouth. Other characteristic features include hoarse voice and pseudomuscular build, and there are distinct skeletal features as well, including an internal notch of the femoral head, internal notch of the second metacarpal, and external notch of the fifth metacarpal. Allelic disorders with overlapping skeletal and joint features include geleophysic dysplasia-2 and the autosomal dominant form of Weill-Marchesani syndrome. Ectopia lentis, familial (OMIM#129600) is caused by heterozygous mutation in the *FBN1* gene (OMIM*134797). Ectopia lentis is defined as an abnormal stretching of the zonular fibers that leads to lens dislocation, resulting in acute or chronic visual impairment. Marfan lipodystrophy syndrome (OMIM#616914) is caused by heterozygous mutation occurring in or affecting exon 64 of the *FBN1* gene (OMIM*134797). The marfanoid-progeroid-lipodystrophy syndrome (MFLS) is characterized by congenital lipodystrophy, premature birth with an accelerated linear growth disproportionate to weight gain, and progeroid appearance with distinct facial features, including proptosis, downslanting palpebral fissures, and retrognathia. Other characteristic features include arachnodactyly, digital hyperextensibility, myopia, dural ectasia, and normal psychomotor development. Marfan syndrome (OMIM#154700), Stiff skin syndrome (OMIM#184900), Weill-Marchesani syndrome 2, dominant (OMIM#608328) and MASS syndrome (OMIM#604308) are caused by heterozygous mutation in the *FBN1* gene (OMIM*134797). A heritable disorder of fibrous connective tissue, Marfan syndrome (MFS) shows striking pleiotropism and clinical variability. The cardinal features occur in 3 systems--skeletal, ocular, and cardiovascular. It shares overlapping features with congenital contractural arachnodactyly, which is caused by mutation in the *FBN2* gene. Stiff skin syndrome (SSKS) is characterized by hard, thick skin, usually over the entire body, which limits joint mobility and causes flexion contractures. Other occasional findings include lipodystrophy and muscle weakness. Patients with similar phenotypes involving stiff skin have been described; see, e.g., familial progressive scleroderma, symmetric lipomatosis, and congenital fascial dystrophy. Weill-Marchesani syndrome is a rare connective tissue disorder characterized by short stature, brachydactyly, joint stiffness, and lens abnormalities. These diseases follow autosomal dominant pattern of inheritance [2].

Variant classification: Based on the evidence, this variant is classified as a likely pathogenic variant. **In this view, clinical correlation and familial segregation analysis are strongly recommended to establish the significance of the finding. If the results do not correlate, additional testing may be considered based on the phenotype observed.**

Recommendations

- Sequencing the variant(s) in the parents and the other affected and unaffected members of the family is recommended to confirm the significance.
- Alternative test is strongly recommended to rule out the deletion/duplication.
- Genetic counselling is advised.

Methodology

DNA extracted from the blood was used to perform whole exome using whole exome capture kit. The targeted libraries were sequenced to a targeted depth of 80 to 100X using GenoLab M sequencing platform. This kit has deep exonic coverage of all the coding regions including the difficult to cover regions. The sequences obtained are aligned to human reference genome (GRCh38.p13) using Sentieon aligner and analyzed using Sentieon for removing duplicates, recalibration and re-alignment of indels. Sentieon DNAscope has been used to call the variants. Detected variants were annotated and filtered using the VarSeq software with the workflow implementing the ACMG guidelines for variant classification. The variants were annotated using 1000 genomes (V2), gnomAD (v4.1.0, v3.1,2.1.1), ClinVar, OMIM, dbSNP, NCBI RefSeq Genes. *In-silico* predictions of the variant was carried out using VS-SIFT, VS-PolyPhen2, PhyloP, GERP++, GeneSplicer, MaxEntScan, NNSplice, PWM Splice Predictor. Only non-synonymous and splice site variants found in the coding regions were used for clinical interpretation. Silent variations that do not result in any change in amino acid in the coding region are not reported.

Sequence data attributes

Total reads generated	9.07 Gb
Data ≥ Q30	95.07%

Genetic test results are reported based on the recommendations of American College of Medical Genetics [1], as described below:

Classification	Interpretation
Pathogenic	A disease-causing variation in a gene which can explain the patients’ symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed
Likely Pathogenic	A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.
Variant of Uncertain Significance	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non- disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.

Disclaimer

- The classification of variants of unknown significance can change over time. Anderson Diagnostics and Labs cannot be held responsible for it.
- Intronic variants, UTR and Promoter region variants are not assessed using this assay.
- This assay has a sensitivity of 70-75% in detecting large deletions/duplications of more than 10 base pairs or copy number variations (CNV). However, it is important to note that any CNVs detected must be confirmed using an alternate method.
- Certain genes may not be covered completely, and few mutations could be missed. Variants not detected by this assay may impact the phenotype.
- The variations have not been validated by Sanger sequencing.
- The above findings and result interpretation was done based on the clinical indication provided at the time of reporting.
- It is also possible that a pathogenic variant is present in a gene that was not selected for analysis and/or interpretation in cases where insufficient phenotypic information is available.
- Genes with pseudogenes, paralog genes and genes with low complexity may have decreased sensitivity and specificity of variant detection and interpretation due to inability of the data and analysis tools to unambiguously determine the origin of the sequence data in such regions.

- Incidental or secondary findings that meet the ACMG guidelines can be given upon request [4].

References

1. Richards, S, et al. Standards and Guidelines for the Interpretation of Sequence Variants: A Joint Consensus Recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. Genetics in medicine: official journal of the American College of Medical Genetics. 17.5 (2015): 405-424.
2. Amberger J, Bocchini CA, Scott AF, Hamosh A. McKusick's Online Mendelian Inheritance in Man (OMIM). Nucleic Acids Res. 2009 Jan;37(Database issue):D793-6. doi: 10.1093/nar/gkn665. Epub 2008 Oct 8.
3. <https://www.ncbi.nlm.nih.gov/clinvar/variation/VCV000920891.5>
4. Miller, David T., et al. "ACMG SF v3. 1 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG)." Genetics in Medicine 24.7 (2022): 1407-1414.

This report has been reviewed and approved by:



Sivasankar.S, Ph.D
Molecular Biologist



Muthukumar.S, Ph.D
Clinical Bioinformatician



Sachin. D.Honguntikar, Ph.D,
Molecular Geneticist



Dr. G. Suriyakumar
Director